

■■ KARNATAKA STATE POLICE

AI-Powered Unified Crime Intelligence & Governance Platform

Datathon 2026 Prototype Pitch Deck

Team Submission | Public Repository & Prototype Presentation

Submitted for: Karnataka State Police Datathon 2026

Deployment: Catalyst / Cloud Ready

GitHub Repository: github.com/aswinpalraj77-coder/project

Tech Stack: React 18, TypeScript, Tailwind, Leaflet

1. Problem Statement & Strategic Vision

Current Police Tech Challenges:

- Fragmented FIR record management across police stations.
- Manual analysis of crime patterns causing delayed response.
- Lack of real-time geospatial intelligence for patrol officers.
- High barrier for citizens to file e-FIRs and track status.
- Information silos between stations hindering link analysis.

Our AI-Driven Solution:

- **Unified Multi-Portal Architecture** (Officer, Citizen, Admin).
- **AI Copilot & FIR Search:** Plain-language queries on crime history.
- **Geospatial Hotspot Analytics** powered by interactive maps.
- **Instant Citizen e-FIR & Emergency SOS** with live GPS dispatch.
- **Criminal Link Graphs** to visualize co-accused networks.

2. Core Modules & Portals Overview

■ Officer Portal	■■ Citizen Portal	■■■ Admin Portal
• AI FIR Natural Language Search • Automated Case Summarizer • Predictive Hotspot Maps • Criminal Link Networks • Shift Handover Copilot	• AI-Guided e-FIR Filing • Real-Time Complaint Tracker • One-Tap Emergency SOS • Cybercrime Reporting Hub • Public Safety Chatbot	• Role-Based Access Control • Station Hierarchy Setup • AI Accuracy & Audit Logs • System Performance Analytics • Token & Cost Monitoring

3. Technology Stack & Deployment Architecture

Layer	Technologies Used	Key Advantages
Frontend UI	React 18, TypeScript, Vite, Tailwind CSS	Lightning fast load time (< 1s), highly responsive UI
Maps & GIS	Leaflet, React-Leaflet, OpenStreetMap	Interactive crime hotspot visualization with zero tile latency
AI & Search	Natural Language Query Parser, Contextual AI	Translates officer prompts into instant structured FIR data
Security & Auth	Role-Based Access Control (RBAC), Supabase Client	Encrypted police data access & audited log tracking
Deployment	Zoho Catalyst / Vercel / Cloud Native	Scalable serverless infrastructure ready for KSP deployment

4. Expected Impact & Submission Summary

■ For Police Officers

- 70% reduction in time spent writing daily crime reports.
- Instant cross-station suspect link detection.
- Data-backed proactive patrol deployment in hotspots.

■ For Citizens

- 24/7 accessible e-FIR filing without station visits.
- Transparent real-time complaint tracking.
- Instant emergency response triggering via SOS dispatch.

■ Submission Links Summary:

- **GitHub Repository:** <https://github.com/aswinpalraj77-coder/project>
- **Prototype Status:** Fully Committed, Built & Production-Ready
- **Hackathon:** Karnataka State Police Datathon 2026